

**Karnataka
Physics- 2018**

General Instructions:

- (1) All the parts are compulsory.**
- (2) Answer without relevant diagram /figure/circuit wherever necessary will not carry any marks.**
- (3) Direct answers to numerical problems without detailed solutions will not carry any marks.**

PART – A

I. Answer all the following question:

1. What is equipotential surface?

Sol: An equipotential surface is a surface with constant value of potential at all points on the surface.

2. Define ‘drift velocity’ of free electrons.

Sol: The average velocity with which the free electrons are drifted in a direction opposite to the applied field is called drift velocity.

3. Give an application of cyclotron.

Sol: Cyclotron is used to accelerate the charged particles or ions to high energies.

4. State Faraday’s law of electromagnetic induction.

Sol: The magnitude of the induced emf in a circuit is equal to the time rate of change of magnetic flux through the circuit.

5. If the peak value of a.c current is 4.24A, what is its root mean square value?

Sol: $I_{rms} = \frac{I_0}{\sqrt{2}} = \frac{4.24}{1.41} = 2.9981A \approx 3A$ ($\because I_0 =$ maximum or peak current)

6. Mention one power +1.5D and -0.50 are kept in contact on their principal axis. What is the effective power of the combination?

Sol: Flux leakage / resistance of the windings / eddy currents / hysteresis loss.

7. Two lenses of power +1.5D and -0.50 are kept in contact on their principal axis. What is the effective power of the combination?

Sol: $P = P_1 + P_2 = 1.5 - 0.5 = 1D$

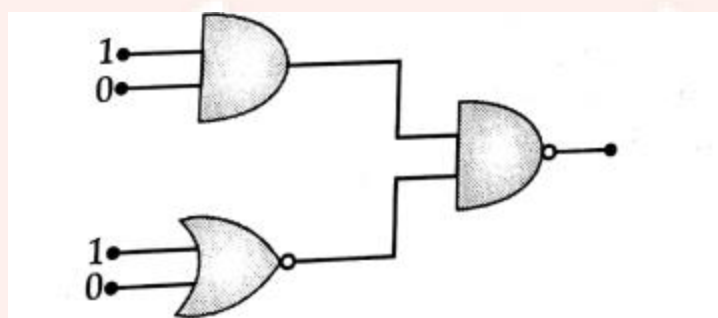
8. The decay of proton to neutron is possible only inside the nucleus. Why?

Sol: The decay of proton to neutron is possible only inside the nucleus because proton has smaller mass than neutron.

9. What is 'depletion region' in a semiconductor diode?

Sol: The space charge region on either side of p-n junction together is known as depletion region.

10.



What is the output of this combination?

Sol: 1

PART-B

II. Answer any five of the following question:

11. Mention any two factors on which the capacitance of a parallel plate capacitor depends.

Sol: Area of the plates / distance between the plates/ permittivity of the medium between the plates.

12. State Kirchhoff's laws of electrical network.

Sol: **Junction rule:** At any junction, the sum of the currents entering the junction is equal to the sum of the currents leaving the junction.

Loop rule: The algebraic sum of the changes in potential around any closed loop involving the resistors and cells in the loop is zero.

13. Define magnetic ‘declination’ and ‘dip’ at a place.

Sol: **Angle of Declination:** At any place on earth, the acute angle between magnetic meridian and the geographical meridian is called the angle of declination.

Angle of Dip: The angle of dip at any place is the angle between earth’s magnetic field intensity B with horizontal in the magnetic meridian at that place.

14. Write the expression for magnetic potential energy of a magnetic dipole kept in a uniform magnetic field and explain the terms.

Sol: $U_m = -\vec{m} \cdot \vec{B}$

U_m = magnetic potential energy, $\vec{m}$ = magnetic dipole moment, B = uniform magnetic field.

15. Give any two applications of X-rays.

Sol:

- (i) X-rays are used as diagnostic tool in medicine.
- (ii) To treat certain forms of cancer.

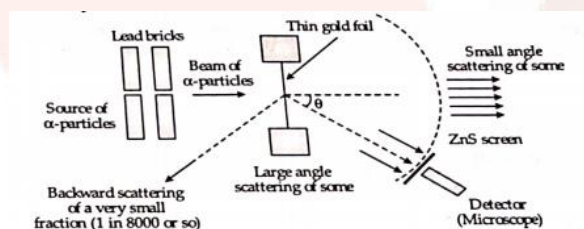
16. What is ‘myopia’? How to rectify it?

Sol: Myopia is a defect in human eye in which light from the distant objects arriving at eye lens get converged at a point before to retina.

17. Draw the diagram representing the schematic arrangement of Geiger-Marsden experimental set up for the alpha particle scattering.

Sol:

Using concave lens to rectify this defect.



18. Write any two characteristics of nuclear forces.

Sol:

- (i) Strongest of all fundamental forces in nature.
- (ii) Charge independent force.
- (iii) Short range force.

- (iv) Saturated force
- (v) Spin dependent force.
- (vi) Non-central force.

PART-C

(III) Answer any five of the following questions:

19. Give three properties of electric charge.

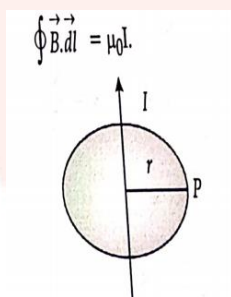
Sol:

- (i) **Additivity of charges:** Charges can be added algebraically.
- (ii) **Charge is conserved:** The total charge of the isolated system is always conserved.
- (iii) **Quantisation of charge:** All free charges are integral multiples of a basic unit of charge 'e'.

20. State Ampere's circuital law and arrive at the expression for the magnetic field near a straight infinite current carrying wire.

Sol: Ampere's circuital law: The integral of the product of magnetic field and length element is equal to μ_0 times the total current passing through the surface.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$



Consider a point at distance 'r' from a straight infinite current carrying wire.

From Ampere's circuital law $\oint \vec{B} \cdot d\vec{l} = B \cdot 2\pi r = \mu_0 I \Leftrightarrow B = \frac{\mu_0 I}{2\pi r}$

21. What is hysteresis? Define the terms ‘coercivity’ and ‘retentivity’ of a ferromagnetic material.

Sol:

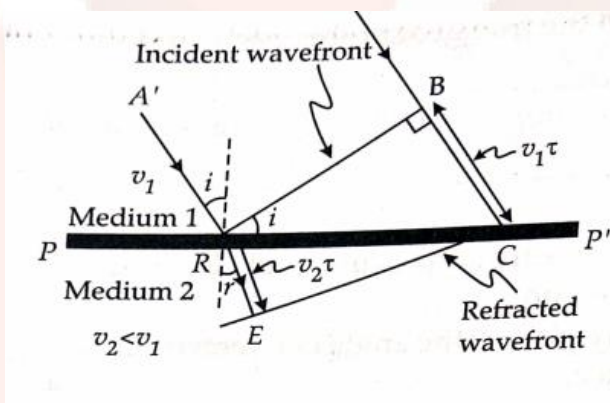
Hysteresis: The phenomenon of lagging behind of magnetic induction (B) with respect to the magnetizing field (H) is called hysteresis.

Coercivity: It's a phenomenon of completely demagnetizing the magnetic material by applying the magnetizing field in the opposite direction.

Retentivity: The property of the magnetic material to retain magnetism even in the absence of the magnetizing field is known as retentivity.

22. Arrive at Snell's law of refraction, using Huygen's principle for refraction of a plane wave.

Sol:



AB = plane wave front

A'A = direction of propagation of wave front

PP' = surface of separation of medium 1 and 2

v_1, v_2 = speeds of light in medium 1 and 2 respectively

i = angle of incidence

r = angle of refraction

τ = time taken by the wave front to travel a distance BC

To determine the shape of the refracted wave front a sphere of radius, $v_2 \tau$ is drawn and

CE = tangent plane if the refracted wave front.

Consider triangles ABC and AEC $\sin i = \frac{BC}{AC} = \frac{v_1 \tau}{AC}$, $\sin r = \frac{AE}{AC} = \frac{v_2 \tau}{AC}$

$$\frac{\sin i}{\sin r} = \frac{v_1 \tau}{v_2 \tau} = \frac{v_1}{v_2} \text{----- (1)}$$

If c = speed of light in vacuum then, $n_1 = \frac{c}{v_1}$, $n_2 = \frac{c}{v_2}$

Equation (1)

$$\Rightarrow \frac{\sin i}{\sin r} = \frac{c}{n_1} \times \frac{n_2}{c} = \frac{n_2}{n_1}$$

$$n_1 \sin i = n_2 \sin r$$

This is the required Snell's Law

23. Write Bohr's postulates for the hydrogen atom model.

Sol:

- (i) An electron in an atom could revolve in certain stable orbits without the emission of radiant energy.
- (ii) Electron revolves around the nucleus only in those orbits for which the angular momentum is some integral multiple of $\frac{h}{2\pi}$ where h = Planck's constant.

$$L = n \cdot \frac{h}{2\pi}$$

- (iii) When an electron makes a transition from one non-radiating orbit to another of lower energy a photon of energy is equal to the energy difference between the two states is emitted. i.e. $h\nu = E_i - E_f$.

24. Derive the expression of the half-life of a radioactive nuclide.

Sol:

We have,

$$N = N_0 e^{-\lambda t}$$

When

$$t = T_{\frac{1}{2}}, N = \frac{N_0}{2}$$

$$\therefore \frac{N_0}{2} = N_0 e^{\lambda T_{\frac{1}{2}}}$$

$$e^{\lambda T_{\frac{1}{2}}} = 2$$

$$\therefore \lambda T_{\frac{1}{2}} = \ln 2 = \log_e 2$$

$$T_{\frac{1}{2}} = \frac{\log_e^2}{\lambda} = \frac{2.303 \log_{10}^2}{\lambda} = \frac{0.693}{\lambda}$$

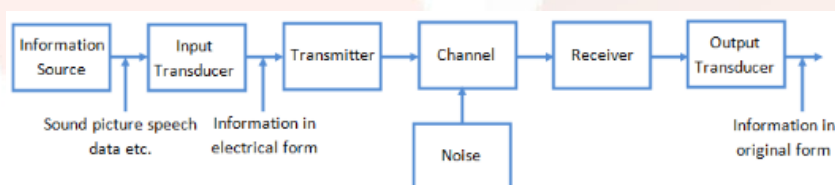
25. Write any three distinctions between p-type and n-type semiconductor.

Sol:

	n-type	p-type
(i)	Obtained by doping pure semiconductor by pentavalent element like As, Sb, P.	Obtained by doping pure semiconductor by trivalent element like Al, B, In.
(ii)	Majority charge carriers are electrons and minority charge carriers are holes.	Majority charge carriers are holes and minority charge carriers are electrons.
(iii)	Donor energy level lies close to the conduction band.	Acceptor energy level lies close to the valence band.
(iv)	$n_e \gg n_h$	$n_h \gg n_e$

26. Draw the block diagram of generalized communication system.

Sol:



PART-D

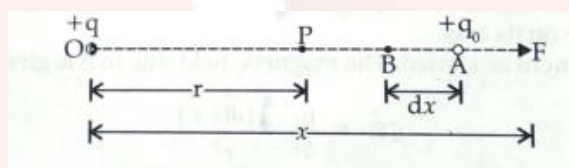
IV. Answer any of the following question:

27. Define electric potential due to a point charge and arrive at the expression for the electric potential at a point due to a point charge.

Sol: Electric potential is defined as the work done in bringing a unit positive test charge from infinity to that point against the direction of the field.

Suppose a point-charge of $+q$ coulomb is situated at a point O in a medium of dielectric constant K. Let P be a point, distant r from O, at which the electric potential is to be determined. For this, we must calculate the work done in bringing a test-charge from infinity to P. Suppose, a test-charge $+q_0$ is placed at point A, distant x from O, and away from P. By Coulomb's law, the magnitude of the electric force $\vec{F}$ acting on q_0 is given by

$$F = \frac{1}{4\pi\epsilon_0 K} \frac{qq_0}{x^2} \text{ newton} \text{----- (1)}$$



The direction of $\vec{F}$ is away from O, if the test charge $+q_0$ is displaced by dx to the point B, the work done

$$dW = F \cdot (-dx)$$

Negative sign is due to the reason that displacement dx is opposite to the direction of force. Hence work done in bringing $+q_0$ from infinity ($x = \infty$) to the point P ($x = r$) is

$$\begin{aligned} W &= -\int_{\infty}^r F dx = -\frac{qq_0}{4\pi\epsilon_0 K} \int_{\infty}^r \frac{1}{x^2} dx \\ &= -\frac{qq_0}{4\pi\epsilon_0 K} \left(-\frac{1}{x} \right)_{\infty}^r = \frac{qq_0}{4\pi\epsilon_0 K} \left(-\frac{1}{r} + \frac{1}{\infty} \right) \\ W &= \frac{1}{4\pi\epsilon_0 K} \frac{qq_0}{r} \end{aligned}$$

Thus, the potential at P

$$V = \frac{W}{q_0} = \frac{1}{4\pi\epsilon_0 K} \frac{q}{r}$$

For vacuum (or air) $K = 1$

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \text{ volt.}$$

28. Obtain the expression for the effective emf and the effective internal resistance of two cells connected in parallel such that the currents are flowing in the same direction.

Sol:

Consider two cells of emf ϵ_1 & ϵ_2 and corresponding internal resistances r_1, r_2 are connected in parallel.

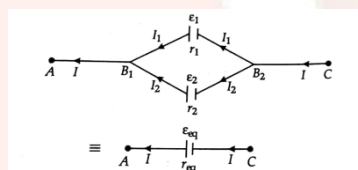
Since as much charge flows in as out, we have

$$I = I_1 + I_2$$

Both the cells at same potential since they are in parallel.

$$V = \epsilon_1 - I_1 r_1 \quad \text{for first cell}$$

$$V = \epsilon_2 - I_2 r_2 \quad \text{for second cell}$$



So,

$$I_1 = \frac{\epsilon_1 - V}{r_1}, I_2 = \frac{\epsilon_2 - V}{r_2}$$

$$I = \frac{\epsilon_1 - V}{r_1} + \frac{\epsilon_2 - V}{r_2} = \frac{\epsilon_1}{r_1} + \frac{\epsilon_2}{r_2} - V \left(\frac{1}{r_1} + \frac{1}{r_2} \right)$$

$$I = \frac{\epsilon_1 r_2 + \epsilon_2 r_1}{r_1 r_2} - V \left(\frac{r_2 + r_1}{r_1 r_2} \right)$$

$$\Rightarrow I(r_1 r_2) = \epsilon_1 r_2 + \epsilon_2 r_1 - V(r_2 + r_1)$$

$$\Rightarrow I \frac{(r_1 r_2)}{(r_1 + r_2)} = \frac{\epsilon_1 r_2 + \epsilon_2 r_1}{(r_1 + r_2)} - V$$

$$\therefore V = \frac{\epsilon_1 r_2 + \epsilon_2 r_1}{(r_1 + r_2)} - V \frac{(r_1 r_2)}{(r_1 + r_2)}$$

$$V = \epsilon_{equivalent} - I r_{equivalent}$$

$$\therefore \epsilon_{eq} = \frac{\epsilon_1 r_2 + \epsilon_2 r_1}{(r_1 + r_2)} \text{ and } r_{eq} = \frac{(r_1 r_2)}{(r_1 + r_2)} \text{ or } \frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2}$$

29. Derive the expression for the magnetic field on the axis of a circular current loop, using Biot-Savart's law.

Sol:

Consider a current loop of radius R carrying a steady current I . let 'P' be a point at a distance 'x' from the center of the current loop on its axis.

Let ' $d\vec{l}$ ' be the current element as shown. The magnetic field due to it is given by Biot-savart's law.

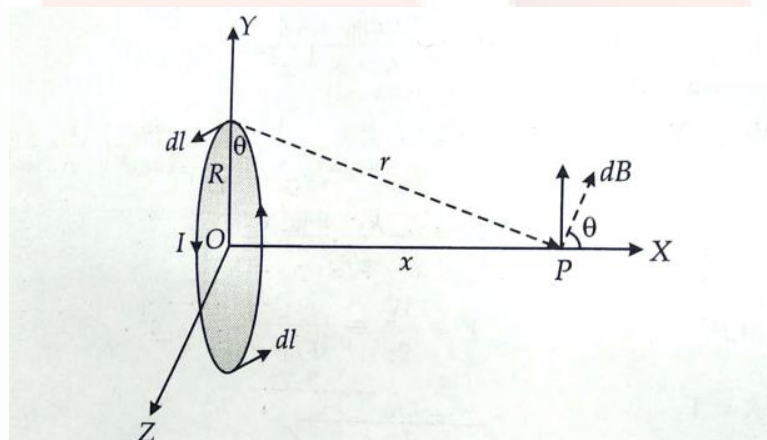
$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I |\vec{dl} \times \vec{r}|}{r^3}$$

But,

$$r^2 = x^2 + R^2$$

The displacement vector $\vec{r}$ from $\vec{dl}$ to the axial point P is in the X-Y plane. Hence

$$|\vec{dl} \times \vec{r}| = r dl$$



$$dB = \frac{\mu_0}{4\pi} \frac{I |\vec{dl} \times \vec{r}|}{r^3} = \frac{\mu_0}{4\pi} \frac{I \cdot dl}{r^2} = \frac{\mu_0}{4\pi} \frac{Idl}{x^2 + R^2}$$

The component of magnetic field along X- direction is $dB_x = dB \cos \theta$

$$\cos \theta = \frac{R}{r} = \frac{R}{(x^2 + R^2)^{\frac{1}{2}}}$$

$$\therefore dB = \frac{\mu_0}{4\pi} \frac{Idl}{r^2} \frac{R}{(x^2 + R^2)^{\frac{1}{2}}} = \frac{\mu_0}{4\pi} \frac{Idl}{(x^2 + R^2)} \frac{R}{(x^2 + R^2)^{\frac{1}{2}}}$$

$$\therefore dB = \frac{\mu_0 IR dl}{4\pi (x^2 + R^2)^{\frac{3}{2}}}$$

Therefore the total magnetic field at P due to the loop

$$B = \frac{\mu_0 IR dl}{4\pi (x^2 + R^2)^{\frac{3}{2}}}$$

$$\therefore \text{As } \int dl = 2\pi R$$

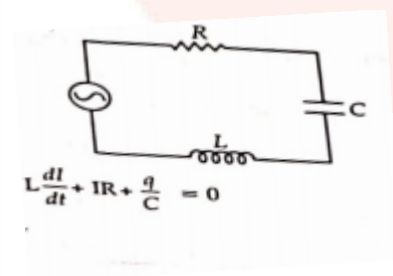
$$B = \frac{\mu_0 IR^2}{2(x^2 + R^2)^{\frac{3}{2}}} = \frac{\mu_0}{2} \frac{IR^2}{(x^2 + R^2)^{\frac{3}{2}}}$$

V. Answer any two of the following questions:

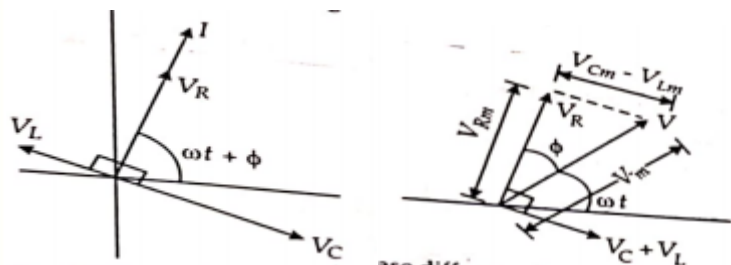
30. Arrive at the expression of the impedance of a series LCR circuit using phasor diagram method and hence write the expression for the current through the circuit.

Sol: Consider a voltage source $V = V_m \sin \omega t$ connected to a series LCR with inductance L, capacitance C and resistance R.

If q is the charge on the capacitor and 'I' the current at any time 't' then from Kirchhoff's rule we have, $V_L + V_R + V_C = 0$



$$L \frac{dl}{dt} + IR + \frac{q}{C} = 0$$



From the diagram $I = I_m \sin(\omega t + \phi)$ where ϕ = phase difference between voltage across the source and the current in the circuit.

$$\vec{V} = \vec{V}_L + \vec{V}_C + \vec{V}_R$$

$\vec{V}_C$ and $\vec{V}_L$ are along same line but in opposite direction.

$\therefore$ The magnitude $|V_{cm} - V_{Lm}|$

$$\therefore V_m^2 = V_{Rm}^2 + [V_{Cm} - V_{Lm}]^2$$

$$V_m^2 = (I_m R)^2 + (I_m X_C - I_m X_L)^2$$

$$V_m^2 = i_m^2 [R^2 + (X_C - X_L)^2]$$

$$Z = \frac{V_m}{I_m} = \sqrt{R^2 + (X_C - X_L)^2}$$

$$Z = \sqrt{R^2 + (X_C - X_L)^2}$$

$$I_m = \frac{V_m}{\sqrt{R^2 + (X_C - X_L)^2}}$$

or,

$$I_m = \frac{V_m}{\sqrt{R^2 + (X_L - X_C)^2}}$$

31. Deduce the relation between n, u, v, Q, R for refraction at a spherical surface, where the symbols have their usual meaning.

Sol:

OM = u = object distance

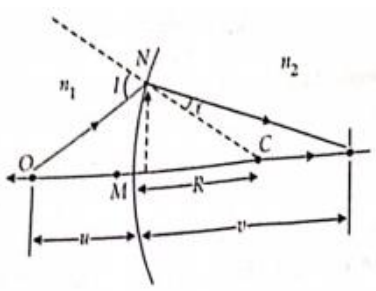
MI = v = image distance

MC = R = radius of curvature

Angle I = angle of incidence

ON = incident ray

NI = refracted ray



NC = normal & n_1, n_2 are the refractive indices

From the figure for small angles

$$\tan \angle NOM = \frac{MN}{OM}, \tan \angle NCM = \frac{MN}{MC}, \tan \angle NCM = \frac{MN}{MI}$$

In the triangle NOC, $\hat{i}$ = exterior angle = sum of the interior opposite angles

$$\angle i = \angle NOC + \angle NCO = \angle NOM + \angle NCM = \frac{MN}{MC} + \frac{MN}{MC}$$

Similarly,

$$\angle NCM = \angle CNI + \angle NIC = \angle CNI + \angle NIM$$

$$\angle r = \angle CNI = \angle MCN - \angle NIM = \frac{MN}{MC} - \frac{MN}{MI}$$

From Snell's law $n_1 \sin i = n_2 \sin r$

For small angles $n_1 i = n_2 r$

Substituting the values of i and r we get,

$$n_1 \left(\frac{MN}{OM} + \frac{MN}{MC} \right) = n_2 \left(\frac{MN}{MC} - \frac{MN}{MI} \right)$$

$$\frac{n_1}{OM} + \frac{n_2}{MI} = \frac{n_2 - n_1}{MC}$$

Applying sign convention,

$$OM = -u, MI = v, MC = R$$

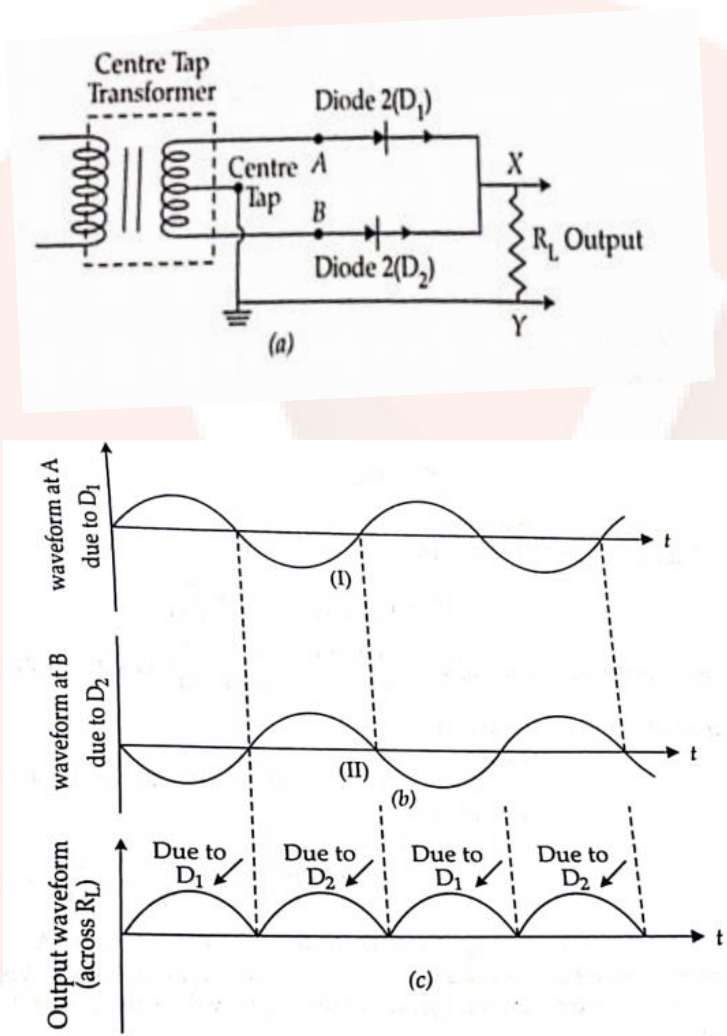
$$\frac{n_1}{v} - \frac{n_2}{u} = \frac{n_2 - n_1}{R}$$

This is the required expression.

32. What is a rectifier? With suitable circuit describe the action of a full wave rectifier by drawing input and output waveforms.

Sol: Rectifier is a device which converts alternating current (ac) into direct current (dc)

For full-wave rectifier:



The circuit connections are as shown in the figure. In the positive half cycle of input AC diode D_1 is forward biased and conducts. In the negative half cycle of the input AC diode D_2 is forward biased and D_1 is reverse biased. D_2 gives output across R_L . The input and output waveforms are as shown. The output is a pulsating DC.

VI. Answer any three of the following questions:

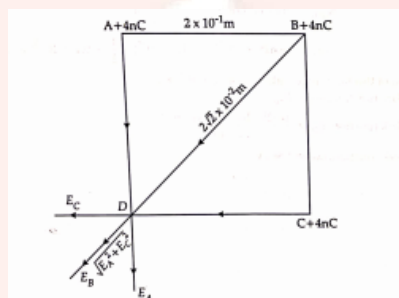
33. Three charges each equal to $+4\text{nC}$ are placed at the three corners of a square of side 2 cm. Find the electric field at the fourth corner.

Sol: The electric field due to charges at A, B & C can be calculated at point D as.

$$E_A = \frac{Q_A}{4\pi\epsilon_0 r^2} = \frac{9 \times 10^9 \times 4 \times 10^{-9}}{(2 \times 10^{-2})^2} = 9 \times 10^4 \text{ NC}^{-1} \text{ along AD}$$

$$E_C = \frac{Q_C}{4\pi\epsilon_0 r^2} = \frac{9 \times 10^9 \times 4 \times 10^{-9}}{(2 \times 10^{-2})^2} = 9 \times 10^4 \text{ NC}^{-1} \text{ along CD}$$

$$E_B = \frac{Q_B}{4\pi\epsilon_0 r^2} = \frac{9 \times 10^9 \times 4 \times 10^{-9}}{(2\sqrt{2} \times 10^{-2})^2} = 4.5 \times 10^4 \text{ NC}^{-1} \text{ along BD}$$



Resultant of E_A and E_C $E_D = \sqrt{E_A^2 + E_C^2} = \sqrt{162 \times 10^8}$ at

$$= 12.73 \times 10^4 \text{ NC}^{-1}$$

Total field at D due to all the charges

$$E_D + E_B = 12.73 \times 10^4 + 4.5 \times 10^4 = 17.23 \times 10^4 \text{ NC}^{-1}.$$

34. 100mg mass of nichrome metal is drawn into a wire of area of cross-section 0.05mm^2 . Calculate the resistance of this wire. Given density of nichrome $8.4 \times 10^3 \text{ kgm}^{-3}$ and resistivity of the material as $1.2 \times 10^{-6} \Omega\text{m}$.

Sol: Since,

$$R = \rho \frac{l}{A} \text{ --- (i)}$$

$$\text{Density (D)} = \frac{\text{mass}(m)}{\text{volume}} = \frac{\text{mass}(m)}{\text{length} \times \text{area}(A)}, \text{length}(l) = \frac{m}{A \times D}$$

$$R = \rho \frac{m}{A \times D} \times \frac{l}{A} = \rho \frac{m}{A^2 D}$$

Substituting the values given, $R = \frac{1.2 \times 10^{-6} \times 100 \times 10^{-6}}{(0.05 \times 10^{-6})^2 \times 8.4 \times 10^3} \Omega = R = 5.7 \Omega$

35. A circular coil of radius 10cm and 25 turns is rotated about its vertical diameter with an angular speed of 40 rad s^{-1} , in a uniform horizontal magnetic field of magnitude $5 \times 10^{-2} \text{ T}$. Calculate the emf induced in the coil. Also find the current in the coil if the resistance of the coil is 15Ω .

Sol: Emf induced in the coil $\varepsilon = nAB\omega \sin(\omega t) = \varepsilon_0 \sin \omega t$ and Area $= \pi r^2$

$$\varepsilon = (25 \times 3.14 \times 10 \times 10^{-2} \times 5 \times 10^{-2} \times 40) \sin(40t)$$

$$\varepsilon = 1.57 \sin 40t$$

$$\varepsilon_0 = 1.57 \text{ volt}$$

$$I = \frac{\varepsilon}{R} = \frac{1.57(\sin 40t)V}{15}, I_{\max} = \frac{\varepsilon_0}{R} = \frac{1.57V}{15\Omega} = 0.1047A = 0.11A$$

Current $I_{\text{inst}} = 0.1047 \sin 40tA$ OR $I_{\max} = 0.11A$

36. In Young's double slit experiment the slits are separated by 0.28 mm and the screen is placed at a distance of 1.4m away from the slits. The distance between the central bright fringe and the fifth dark fringe width if the screen is moved 0.4m towards the slits, for the same experimental set up.

Sol: Fringe width $\beta = \frac{\lambda D}{d}$

Given,

The distance of the fifth dark fringe from the central bright fringe $= 1.35 \text{ cm}$

$$\beta = \left(m - \frac{1}{2}\right) \frac{D\lambda}{d}$$

Wavelength of light used $\lambda = 600 \times 10^{-9} \text{ m}$, $m = 5$

Taking distance between the slits and the screen as $1 \text{ m} \Rightarrow \beta = \frac{600 \times 10^{-9} \times 9 \times 1}{2 \times 1.35 \times 10^{-2} \times 0.28 \times 10^{-3}}$

Calculation of fringe width $7.142 \times 10^{-3} \text{ m}$.

37. Light of frequency $8.41 \times 10^{14} \text{ Hz}$ is incident on a metal surface. Electrons with their maximum speed of $7.5 \times 10^5 \text{ ms}^{-1}$ are ejected from the surface. Calculate the threshold frequency for photo emission of electrons. Also find the work function of the metal in electron volt (eV). Given Planck's constant $h = 6.625 \times 10^{-34} \text{ Js}$ and mass of the electron $9.1 \times 10^{-31} \text{ Kg}$.

Sol:

$$K_{\max} = hv - \omega_0 \text{ OR } hv = hv_0 + \frac{1}{2}mv^2$$

$$hv_0 = (6.625 \times 10^{-34} \times 8.41 \times 10^{14}) - \frac{1}{2} \times 9.1 \times 10^{-31} \times (7.5 \times 10^5)^2$$

Calculation of threshold frequency $\nu_0 = 4.54 \times 10^{14} \text{ Hz}$

Work function, $\omega_0 = hv_0$

$$\text{Work function, } \omega_0 = \frac{6.25 \times 10^{-34} \times 4.54 \times 10^{14}}{1.6 \times 10^{-19}}$$

Work function $\omega_0 = 1.88 \text{ eV}$